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Article Title:

A High-Resolution Finite-Difference Framework for Steady Continuation and Finite-Window Unsteady Dynamics up to $Re = 100,000$ in 2D Lid-Driven Cavity Flows

Article Classification: Research Paper

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- **Open-Source Code Repository:** <https://github.com/KartikeyaGangwar/lid-driven-cavity-cfd>
- **Permanent Dataset Archive DOI:** [10.5281/zenodo.18312938](https://doi.org/10.5281/zenodo.18312938)

Structured Abstract (223 words):

- **Purpose:** To resolve the computational bottlenecks of iterative residual drift, artificial numerical dissipation, and high-Reynolds divergence in the canonical 2D incompressible lid-driven cavity problem from $Re = 100$ up to $Re = 100,000$.
- **Design/methodology/approach:** A finite-difference framework solves the streamfunction–vorticity equations by coupling an exact $\mathcal{O}(N^2 \log N)$ Discrete Sine Transform (DST-I) Poisson solver with vectorized Alternating Direction Implicit (ADI) marching. Pure second-order central differencing (zero artificial viscosity, $\nu_{\text{art}} = 0$) is preserved for $Re \leq 50,000$. At $Re = 100,000$ (cell Péclet > 195), an interior directional hybrid model regularizes driven corner singularities with explicit numerical dissipation mapping.
- **Findings:** Stationary continuation on Cartesian meshes up to 1025×1025 captures multi-scale corner eddy cascades and demonstrates qualitative alignment with Batchelor’s core homogenization hypothesis (vortex center $x_c \rightarrow 0.50000$, plateau core vorticity $\omega_0 \approx -2.54$). Spatial discretization achieves ASME V&V 20 grid convergence ($GCI = 0.48\%$). Time-accurate marching on 513×513 meshes resolves coherent vortex shedding at $Re = 10,000$ ($St = 0.6249$), modulated dynamics at $Re = 15,000$ ($St = 0.5364$), and finite-window shedding modes up to $Re = 100,000$.
- **Originality/value:** This study provides an open-source, highly scalable spectral-accelerated FDM framework, extends verified benchmark solutions to extreme Reynolds numbers ($Re = 100,000$), and openly releases thirty validated full-field datasets.

Keywords (6):

Lid-driven cavity flow, Finite difference method, Streamfunction–vorticity, Fast Poisson solver, Unsteady vortex shedding, Batchelor theorem.

CRedit Authorship Contribution Statement:

Kartikeya Gangwar: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project administration.

Declaration of Competing Interest:

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability Statement:

The complete source code, numerical datasets (flow fields, benchmark profiles, verification metrics), and telemetry files are openly available under an MIT license on GitHub at <https://github.com/KartikeyaGangwar/lid-driven-cavity-cfd> and archived permanently on Zenodo with DOI: [10.5281/zenodo.18312938](https://doi.org/10.5281/zenodo.18312938).

Declaration of Generative AI:

During the preparation of this work, the author used AI-assisted technologies solely for editorial consistency, grammar verification, and scientific language polishing. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content and conclusions of the published article.